Question:

A 34-year-old poultry worker presents to his physician with a sore throat and a non-productive cough for 2 weeks. His cough is associated with fever. The vital signs include: blood pressure 120/80 mm Hg, heart rate 67/min, respiratory rate 18/min, and temperature 37.6°C (98.0°F). Physical examination shows oropharyngeal erythema and scattered, moist rales on lung auscultation. The patient’s X-ray demonstrates patchy reticular opacities in the perihilar regions of both lungs. After some additional tests, he is diagnosed with community-acquired pneumonia and is treated with cephalexin with modest improvement. Which of the following best describes the immune response elicited by the pathogen that is causing this patient’s condition?

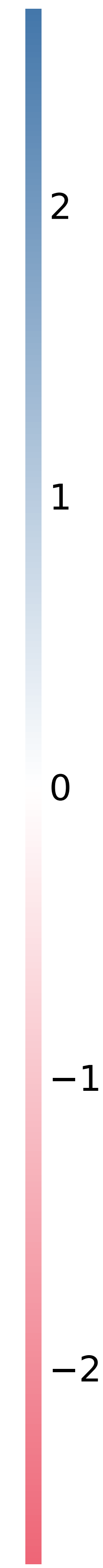
Answer Choices:

A. It activates TLR5 on the surface of macrophages.

B. It primarily induces the Th1-cell response.

C. Peptidoglycan is its major antigen that induces an immune response.

D. This pathogen evades the immune response by encapsulation.



Reasoning + Answer:

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- The patient presents with symptoms of community-acquired pneumonia, which typically affects individuals who have not been hospitalized recently.
- The pathogen causing the pneumonia may be Gram-positive, based on the effectiveness of the antibiotic cephalexin (an antibiotic for Gram-positive bacteria).
- Cephalosporins like cephalexin are not effective against encapsulated bacteria and those that primarily induce a Th1 response.
- Gram-positive bacteria often use peptidoglycan in their cell wall, and it is a major antigen that can induce an immune response.

Given these points, the best answer appears to be that peptidoglycan is the major antigen that induces an immune response, as this aligns with common pathogens causing such symptoms and the effectiveness of the antibiotic used.

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<answer>C. Peptidoglycan is its major antigen that induces an immune response.</answer>